

Yiming Ma

Research Associate @ KCL

PROFILE

Computer Vision & Multimodal Learning researcher with strong publication record (7 papers, 6 first-author, ~140 citations) and industry collaboration experience. Experienced in crowd counting, multi-view multimodal driver monitoring. Seeking postdoctoral roles advancing multimodal foundational models.

PUBLICATIONS & PREPRINTS

- **2026 - ECCVW:** *TACIT: Task-Aware Counterfactual Image Translation for Medical Domain Generalization*.
 - **Motivation:** Existing translation losses cannot ensure that synthetic medical images preserve task labels while matching the target domain.
 - **Method:** Introduces TACIT, which generates and filters counterfactual images using leakage-aware task and domain validators.
 - **Results:** Achieves 96.04% test accuracy on Camelyon17-WILDS and improves PI-CAI accuracy from 69.71% to 75.14%.
- **2025 - IEEE TIP (under review):** *ZIP: Scalable Crowd Counting via Zero-Inflated Poisson Modeling*.
 - **Motivation:** Ground-truth blockwise density maps are highly sparse (>95% zero blocks); MSE poorly models blockwise counts.
 - **Method:** Proposes ZIP that models blockwise counts with a Zero-Inflated Poisson.
 - **Results:** SOTA results on ShanghaiTech A & B, UCF-QNRF and NWPU with models ranging from <1M to ~100M parameters.
- **2025 - ICME:** *CLIP-EBC: CLIP Can Count Accurately through Enhanced Blockwise Classification*.
 - **Motivation:** CLIP in regression-style dense prediction is underexplored; existing blockwise counting models rely on Gaussian-smoothed labels, causing ambiguity.
 - **Method:** Introduces EBC based on integer-valued bins; proposes CLIP-EBC, the first purely CLIP-based model that achieves accurate crowd counting results.
 - **Results:** Large gains over prior blockwise methods; SOTA on NWPU-Test (MAE 58.2).
- **2025 - ICME (co-author):** *Interact with me: Joint Egocentric Forecasting of Intent to Interact, Attitude and Social Actions*.
 - **Motivation:** Assistive/AR systems need early forecasts of social intent and actions from egocentric video.
 - **Method:** A joint forecasting framework that shares an egocentric encoder and learns multi-task heads for intent, attitude, and social actions with temporal modeling.
 - **Results:** Benchmarked on standard egocentric datasets, with quantitative and qualitative analyses showing the benefits of joint modeling for consistent forecasts.
- **2023 - CVPRW:** *Robust Multiview Multimodal Driver Monitoring System Using Masked Multi-Head Self-Attention*.
 - **Motivation:** Real-world DMS must be robust to view/modality collapse and occlusions.
 - **Method:** Masked multi-head self-attention to fuse Top/Front × IR/Depth streams; supervised contrastive learning and robustness regularization.

- **Results:** On DAD, four-stream fusion reaches AUC-ROC 97.0% / mAP 97.8%, outperforming decision-level fusion and alternative feature-level baselines.
- **2022 - ICIP:** *FusionCount: Efficient Crowd Counting via Multiscale Feature Fusion.*
 - **Motivation:** Encoder-decoder counters underuse low-level features and add heavy multiscale modules.
 - **Method:** Contrast-aware group-wise fusion of encoder features plus a dual-branch channel-reduction decoder (1×1 + dilated conv).
 - **Results:** ShanghaiTech-B MAE 6.9 / RMSE 11.8 with ~815GFLOPs, surpassing or matching VGG-based peers (CSRNet, CAN, BL, DM-Count) at lower compute.
- **2022 - arXiv:** *Real-Time Driver Monitoring Systems through Modality and View Analysis.*
 - **Motivation:** 3D CNNs are costly while adjacent frames in in-cabin video are highly similar.
 - **Method:** Image-level DMS with 2D encoders and (feature/decision) fusion across views/modalities; evidence for dropping explicit temporal modeling; open-set handling.
 - **Results:** AUC-ROC 95.6 / Accuracy 92.4% on DAD with 243M-1.01GFLOPs.

EXPERIENCE

RESEARCH ASSOCIATE, KING'S COLLEGE LONDON; LONDON, UK – 2026-PRESENT

- **Multimodal fingerprint:** Develop patient-level multi-modal fingerprints by integrating longitudinal prostate MRI with clinicopathological variables, demographic information, and radiology reports using foundation models and self-/semi-supervised learning.
- **Longitudinal prediction:** Build longitudinal models to predict prostate cancer progression and support personalised active-surveillance strategies, addressing incomplete follow-up data, missing modalities, domain shift, and external validation.

RESEARCH ASSISTANT, UNIVERSITY OF WARWICK; COVENTRY, UK – 2022-2023

- **Data curation:** Refined and extended DAD annotations, adding 9 non-driving-related activities; prepared data for robust benchmarking.
- **Multiview multimodal fusion:** Designed a multi-view multimodal driver monitoring system based on masked multi-head self attention; improved AUC-ROC from 88% to 97% on DAD and increased robustness to view/modality collapse.

TEACHING ASSISTANT, UNIVERSITY OF WARWICK; COVENTRY, UK – 2023

- **Lab sessions:** Assisted delivery of an undergraduate Python & Introductory ML module; led labs and tutorials guiding students to implement regression, classification, and neural networks in Python.
- **Tutoring:** Provided one-to-one and small-group academic support, clarifying core programming/ML concepts and troubleshooting code and experiment design.

EDUCATION

PHD IN MATHEMATICS OF SYSTEMS, UNIVERSITY OF WARWICK; COVENTRY, UK – 2021 - 2025 SEPT

- Supervisors: Prof. [Victor Sanchez](#) & Dr. [Tanaya Guha](#).

MSC IN MATHEMATICS OF SYSTEMS UNIVERSITY OF WARWICK (DISTINCTION); COVENTRY, UK – 2020 - 2021;

- Research: Prediction of Oestrus Intervals for Guide Dogs (supervisor: Prof. [Colm Connaughton](#)); Digital Twins of Urban Crowds (supervisors: Prof. [Theo Damoulas](#), Prof. [Victor Sanchez](#) & Dr. [Tanaya Guha](#)).

BSC IN MATHEMATICS AND APPLIED MATHEMATICS, SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY (GPA: 3.89/4.00); SHENZHEN, CHINA – 2016 - 2020

SERVICE

Reviewer for TNNLS, TMM, SPL, ECCV, ACM MM, CVPRW, ICME, WACV, BMVC, ICIP.

SKILLS

- **Deep Learning Concepts:** Attention Mechanism, ViT, Prompt Tuning, CLIP, Contrastive Learning, Multimodal Alignment, Multimodal Fusion.
- **PyTorch:** TIMM, OpenCLIP, Transformers, TensorBoard, Optuna.
- **Python:** NumPy, SciPy, Scikit-learn, Pandas, OpenCV, Matplotlib / Seaborn.
- **Maths & Stats:** Probability Theory, Statistical Inference, Optimization, Stochastic Processes, Time Series Modeling, Survival Analysis, Computational Statistics, Real / Complex / Functional / Fourier Analysis, Measure Theory.
- **Development & Tools:** SSH, Git, Linux, LaTeX, Markdown, MS Word.
- **Languages:** Mandarin Chinese (native), English (IELTS: 8.0/9.0).